

Design Proposal for a Silicon Photonic Mach-Zehnder Interferometer for an Electron Beam Lithography Fabrication Process

Karina Soriano
Edx: @rinasoriano
GitHub: mkla-soriano
Psionic Inc.

I. INTRODUCTION

The Mach-Zehnder Interferometer (MZI) has been utilized over time to govern relative optical phase shifts variations in two different beams caused by one source. In integrated photonic platforms, it has been used for thermo-optic and electro-optic modulators. In this proposal, we will design a variation of imbalanced MZIs with different delay times and report their simulated results.

II. THEORY

Let's consider a splitter called a y-branch. It has a input intensity, I_i , and has an electric field, E_i . The beam is equally separated into the two branches of the y-branch. Each of the outputs has the following intensity and electric field respectively: $I_1 = I_2 = \frac{I_i}{2}$ and $E_1 = E_2 = \frac{E_i}{\sqrt{2}}$. This is due to $I \propto |E|^2$. The same equations apply for the combiner, in which the light input in one of the waveguides, E_i , is evenly split between the fundamental and higher order modes, also known as radiation modes, of the waveguide. Therefore, the output of the light at the output port will be $E_{o1} = \frac{E_1}{\sqrt{2}}$ or $E_{o2} = \frac{E_2}{\sqrt{2}}$. The output field, with two inputs, becomes a vector summation of the input fields. It is divided by $\sqrt{2}$, which becomes $E_o = \frac{E_1 + E_2}{\sqrt{2}}$. The electric field for the plane wave then becomes $E_0 e^{i(\omega t - \beta z)}$, with propagation constant of the beam in the arms being $\beta_1 = \frac{2\pi n_1}{\lambda}$ and $\beta_2 = \frac{2\pi n_2}{\lambda}$, in which n is the index of refraction. The lengths in the waveguides are L_1 and $L_2 = L_1 + \Delta L$. At the end of the waveguides, the outputs are:

$$E_{o1} = E_1 e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} \quad (1)$$

$$E_{o2} = E_2 e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} = \frac{E_i}{\sqrt{2}} e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \quad (2)$$

Once the light is combined in the y-branch, the output from the combiner is as follows:

$$E_o = \frac{1}{\sqrt{2}} (E_{o1} + E_{o2}) = \frac{E_i}{2} (e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}) \quad (3)$$

The intensity becomes as follows:

$$I_o = \frac{I_i}{4} = |e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}|^2 \quad (4)$$

For an imbalanced interferometer, where the waveguides are identical, the intensity is as follows:

$$I_o = \frac{I_i}{2} [1 + \cos(\beta \Delta L)] \quad (5)$$

III. MODELING AND SIMULATION

A. Waveguide

For this particular design, we will be using a strip waveguide with the following dimensions: 500nm for the width and 220nm for the height. The following images are that of the simulated electric field intensity transverse electric (TE) and transverse magnetic (TM) mode in the waveguide created in Lumerical MODE.

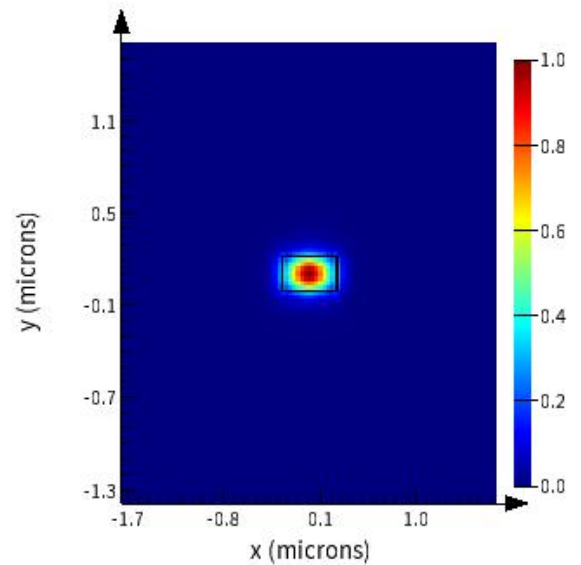


Fig. 1. TE mode in the waveguide.

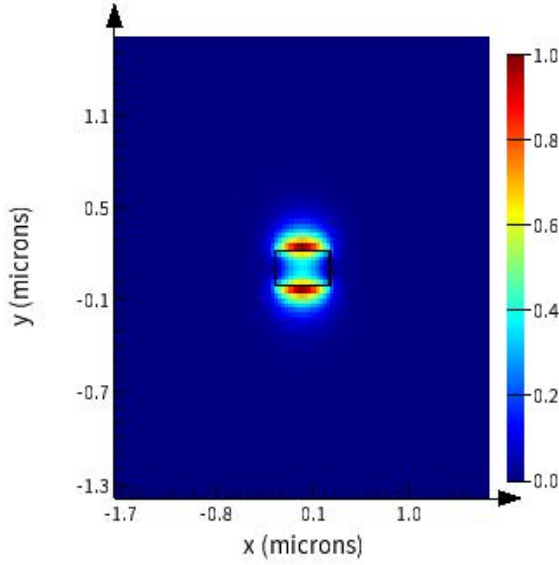


Fig. 2. TM mode in the waveguide.

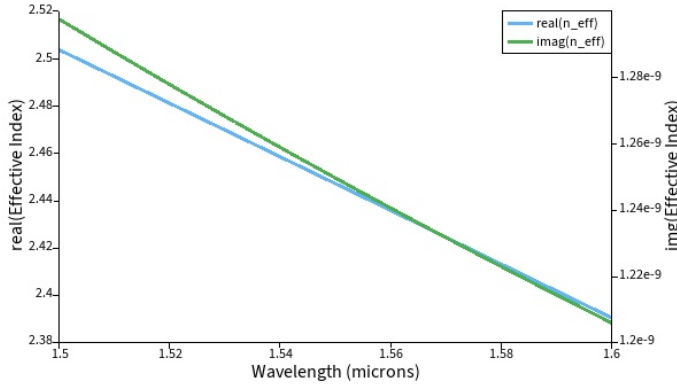


Fig. 3. Effective index (n_{eff}) of waveguide vs wavelength.

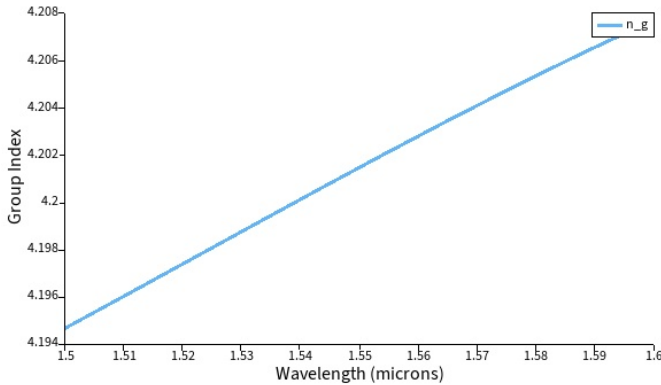


Fig. 4. Group index (n_g) of waveguide vs wavelength.

Fig. 1 illustrates the electric field intensity of TE mode in the waveguide, while Fig. 2 illustrates the electric field intensity TM mode. Figures 3 and 4 illustrates the effective index (n_{eff}) of waveguide vs wavelength and group index (n_g) of waveguide vs wavelength respectively. We can deduce that n_{eff} decreases with wavelength while we see an increase for n_g .

We can use a Taylor expansion around the center wavelength in the following form:

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \quad (6)$$

Using Lumerical Interconnect, the compact model of this particular waveguide can be calculated as follows:

$$n_{eff}(\lambda) = 2.44 - 1.13(\lambda - \lambda_0) - 0.04(\lambda - \lambda_0)^2 \quad (7)$$

B. Mach-Zehnder Interferometer

In the Theory section, we can acquire the transfer function of the MZI as follows:

$$T_{MZI}(\lambda) = \frac{1}{2}[1 + \cos(\beta\Delta L)] \quad (8)$$

The interferometer's free spectral range (FSR) can be written as $FSR = \frac{\lambda^2}{n_g\Delta L}$. Different ΔL have the following FSR, shown in Table 1 below:

$\Delta L(\mu m)$	$FSR(nm)$
125	4.59649
225	2.49793
325	1.79862
425	1.40128
525	1.10065
625	0.90030

TABLE I
FSR WITH VARYING ΔL

Using Lumerical INTERCONNECT, we simulate the MZI to generate the spectrum of the interferometer as seen on Figure 5 below. TE mode was selected with a ΔL of $425\mu m$.

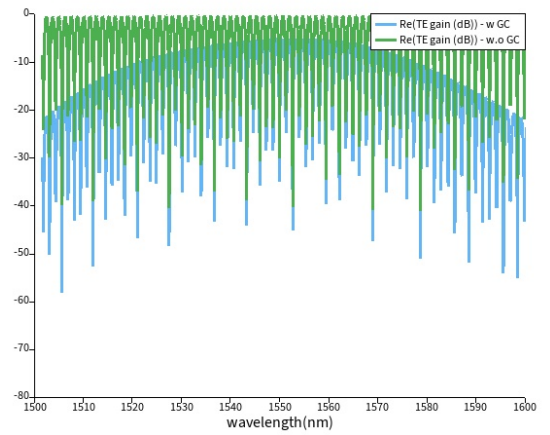


Fig. 5. MZI Spectrum with a ΔL of $425\mu m$

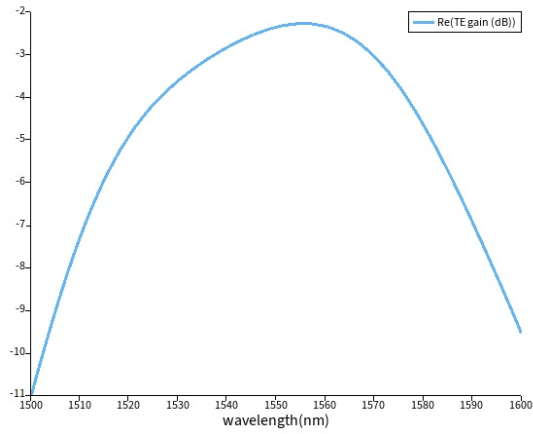


Fig. 6. Insertion loss in each of the two grating couplers.

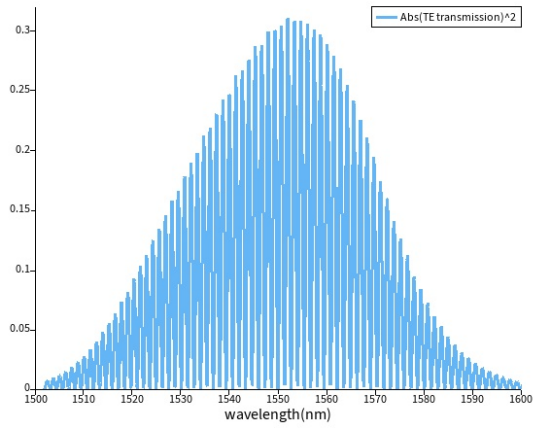


Fig. 7. MZI Transmission with two grating couplers.

Figure 6 indicates the insertion loss in each of the two grating couplers in the MZI, while Figure 7 illustrates the transmission with the two grating couplers.

IV. FABRICATION

To be completed later.

V. EXPERIMENT DATA

To be completed later.

VI. ANALYSIS

To be completed later.

VII. EXPERIMENT DATA

To be completed later.

VIII. CONCLUSION

To be completed later.

REFERENCES

- [1] L. Chrostowski, M. Hochberg, in "Silicon Photonics Design", Cambridge University Press, 2015